

Sci-kit Learn Recreation and Implementation

CSE299-sec-5

Group-5

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Week 4 ML Library Implementation - Project Update Report

Key Accomplishments

Core Clustering Algorithms Implemented: K-Means clustering with k-means++ initialization and convergence detection; Agglomerative Hierarchical Clustering with multiple linkage methods (Ward, complete, average, single); DBSCAN density-based clustering with configurable eps and min_samples parameters and noise point detection. **Enhanced Base Class Architecture:** Extended BaseClusterer with clustering-specific attributes including cluster centers, labels, and cluster count tracking; improved BaseTransformer with dimension tracking; enhanced BaseModel with robust input validation and consistent API design. **Comprehensive Clustering Evaluation System:** Implemented Silhouette Score for cluster coherence measurement; Calinski-Harabasz Score for variance ratio assessment; Davies-Bouldin Score for cluster separation evaluation; Within-Cluster Sum of Squares (WCSS) for inertia calculation; Adjusted Rand Index for comparing clustering results. **Advanced CLI Integration:** Extended command-line interface with clustering-specific parameters including n_clusters, eps, min_samples, linkage methods, and convergence tolerances; automatic task type detection distinguishing clustering from classification and regression; comprehensive parameter validation and error handling. **Robust Data Processing Pipeline:** Enhanced preprocessing integration with full dataset handling for unsupervised learning; improved data validation supporting clustering workflows without target labels; standardized scaling integration ensuring optimal clustering performance. **Quality Assurance and Testing:** Systematic debugging and resolution of numpy import issues and module path problems; comprehensive testing framework covering all clustering algorithms across multiple datasets (Iris, Wine, Digits); performance validation ensuring meaningful results with appropriate evaluation metrics. **Technical Infrastructure Enhancements:** Modular clustering metrics system with independent evaluation functions; enhanced logging and debugging capabilities; improved results management with clustering-specific output formatting including cluster labels, centroids, and performance metrics; standardized error handling across all clustering components.

